COMPUTER GRAPHICS

1. what is the function of a frame buffer?

A frame buffer is a memory area in a computer used to store pixel data for an image before it is displayed on the screen.

2. how do interactive graphics displays work?

Interactive graphics displays allow the user to communicate directly with the computer graphics system using input devices such as keyboard, mouse, or touch.

Interactive graphics displays work by accepting user input, processing it, updating the frame buffer, and refreshing the screen to provide real-time visual feedback.

3. give one example of an emissive display technology and one example of non emissive display technology.

Emissive display example: Cathode Ray Tube (CRT)
Non-emissive display example: Liquid Crystal Display (LCD)

4. Difference Between Flood Fill and Boundary Fill Algorithm

Feature	**Flood Fill Algorithm**	**Boundary Fill Algorithm**
Basic Idea	Fills area until a **specified interior color** is replaced	Fills area until a **boundary color** is reached
Input Required	Starting point and **fill color + old color**	Starting point, **fill color + boundary color**
Stopping Condition	Stops when pixel color ≠ old color	Stops when boundary color is encountered
Dependency	Depends on **interior color**	Depends on **boundary color**
Area Requirement	Region must have **uniform interior color**	Region must be enclosed by a **distinct boundary**
Risk of Leakage	Less leakage if interior color uniform	May leak if boundary not continuous
Comparison Used	Compare pixel with **old color**	Compare pixel with **boundary color**
Usage	Filling multi-colored boundaries with same interior	Filling regions with clearly defined edges

5. difference between point clipping and line clipping table format

Feature	Point Clipping	Line Clipping
Definition	Determines whether a **point** lies inside or outside the clipping window	Determines which **portion of a line** lies inside the clipping window
Object Handled	Single point	Line segment (two endpoints)
Complexity	Simple	More complex
Calculation	Only checks coordinates of the point	Calculates intersections with window boundaries
Output	Point is either **accepted or rejected**	Line may be **fully accepted, fully rejected, or partially accepted**
Condition	Check if point lies within xmin, xmax, ymin, ymax	Check endpoints and clip against boundaries
Processing Time	Less	More
Example Use	Checking cursor location	Clipping lines in drawing applications

6. define homogeneous coordinates and their importance in 3D transformation

Homogeneous coordinates are a way of representing points in 3D space by adding an extra coordinate.
A 3D point $(x, y, z)$is written in homogeneous form as (x, y, z, w), where usually $w = 1$.

Importance:

- Allows all transformations to be written using **4 × 4 matrices**.
- Multiple transformations (translation, rotation, scaling) can be combined into a **single matrix**.
- All transformations use the **same matrix multiplication approach**.

7. write the transformation matrix for 3d scaling

$$\begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(x′,y′,z′)=(Sxx,Syy,Szz)

8. what is composite transformation?

A **composite transformation** is the **combination of two or more transformations** (such as translation, rotation, scaling) applied **in sequence** to an object.

Instead of applying each transformation separately, they are **combined into a single transformation matrix** and applied once.

9. how does the scan-line fill algorithm function?

The scan-line fill algorithm fills a polygon by scanning one horizontal line at a time, finding intersection points with polygon edges, sorting them, and filling pixels between each pair of intersections until the entire region is filled.

1. explain the function of pointing and positioning devices

Pointing Devices 🖱️

Pointing devices are input devices used to **select objects or commands** on the screen. They allow the user to point to icons, menu items, or graphical elements and perform actions like clicking, dragging, and selecting. These devices mainly control the cursor movement. Examples include mouse, light pen, and trackball. They are widely used in interactive graphics for object selection and manipulation.

Positioning Devices 🎯

Positioning devices are used to **specify exact coordinates** on the screen for placing graphical objects. They help in accurately locating points, drawing shapes, and controlling object placement. These devices provide precise positional input to the graphics system. Examples include digitizer, joystick, and touch screen. They are useful for drawing and designing applications.

2. describe the role of focusing system in CRT

The **focusing system** in a **CRT (Cathode Ray Tube)** is used to **control and concentrate the electron beam** so that it strikes a **small, sharp spot** on the phosphor screen.

Functions:

- 🎯 **Narrows the electron beam** to improve clarity
- 🖼️ **Produces sharp and well-defined images**
- ⚙️ **Reduces spreading of electrons** while traveling
- 📏 **Maintains uniform focus** across the screen

3. if the resolution of an image increases while maintaining the same display size , how will it affect the PPI and aspect ratio

PPI (Pixels Per Inch) ☑
When resolution increases while the display size remains the same, **PPI increases**.
This means **more pixels are packed into the same physical area**, resulting in **sharper and clearer images**.

Aspect Ratio 📏
The **aspect ratio remains unchanged** *if* both width and height resolution increase proportionally.
If they increase unevenly, the aspect ratio **will change**.

4. write the equation of an ellipse centered at the origin

The equation of an **ellipse centered at the origin** is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Where:

- a= semi-major axis length
- b= semi-minor axis length
- Center of ellipse = $(0, 0)$

5. difference between clipping and windowing in computer graphics

Feature	**Clipping**	**Windowing**
Definition	Removes parts of objects outside the viewing area	Selects a portion of the world coordinate for display
Purpose	Display only the visible part of objects	Define area of interest
Operation	Eliminates unwanted portions	Chooses region to be viewed
Effect on Objects	Parts outside region are discarded	Entire selected region is mapped to viewport
Processing Stage	Done after window is defined	Done before viewport mapping
Output	Partial object may remain	Whole window area is displayed
Example	Cutting a line at window boundary	Selecting a zoomed area

6. what is a polygon? what are the concave and convex polygons?

A **polygon** is a closed two-dimensional figure formed by **joining three or more straight line segments** end-to-end. The line segments are called edges and their intersections are called vertices.

A **convex polygon** is a polygon in which **all interior angles are less than 180°**, and any line drawn between two points inside the polygon lies completely inside it.

A **concave polygon** is a polygon in which **at least one interior angle is greater than 180°**.

1. what is computer graphics? describe its application in computer aided design and image processing.

Computer graphics is the branch of computer science that deals with the **creation, storage, manipulation, and display of images and graphical objects** using computers. It allows users to generate pictures, drawings, animations, and visual representations of data.

Application in Computer Aided Design (CAD) 📐

- Used to design mechanical parts, buildings, and electronic circuits
- Enables **2D and 3D modeling** of objects
- Allows easy **modification and visualization** of designs

Application in Image Processing 🖼

- Enhances image quality (brightness, contrast, filtering)
- Used in **medical imaging** like X-rays and MRI
- Supports **pattern recognition and object detection**

2. difference between random scan and raster scan

Feature	**Random Scan Display**	**Raster Scan Display**
Drawing Method	Draws images using **lines directly**	Draws images using **pixels line by line**
Beam Movement	Electron beam moves only where picture is drawn	Electron beam scans entire screen row by row
Resolution	Depends on line drawing accuracy	Fixed resolution (pixel based)
Memory Requirement	Requires less memory	Requires more memory (frame buffer)
Picture Quality	Best for line drawings	Best for filled images and shading
Refresh	Refreshes only drawn lines	Refreshes whole screen
Cost	Expensive	Less expensive
Example	Vector displays	TV and modern monitors

3. what is viewing pipeline? explain the process of window to viewport transformation

The **viewing pipeline** is the sequence of steps used to convert objects from **world coordinates to screen coordinates** for display. It transforms a scene into the final image on the display device.

Window-to-Viewport Transformation

Window-to-viewport transformation converts a selected **window** (in world coordinates) into a **viewport** (display screen coordinates).

Step-by-Step Process

Step 1: Define Window
Select the region in world coordinates:

- x_{wmin}, x_{wmax}
- y_{wmin}, y_{wmax}

Step 2: Define Viewport
Select the area on display screen:

- x_{vmin}, x_{vmax}
- y_{vmin}, y_{vmax}

Step 3: Compute Scaling Factors

$$S_x = \frac{x_{vmax} - x_{vmin}}{x_{wmax} - x_{wmin}}$$
$$S_y = \frac{y_{vmax} - y_{vmin}}{y_{wmax} - y_{wmin}}$$

Step 4: Apply Transformation
Map each point from window to viewport:

$$x_v = x_{vmin} + (x_w - x_{wmin}) \frac{x_{vmax} - x_{vmin}}{x_{wmax} - x_{wmin}}$$

$$y_v = y_{vmin} + (y_w - y_{wmin}) \frac{y_{vmax} - y_{vmin}}{y_{wmax} - y_{wmin}}$$

Step 5: Display Result

The transformed coordinates $(x_v, y_v)$are drawn inside the viewport on the screen.